

NEXORA



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Next-generation Explainable Route Optimization & Retrieval Assistant

AI-Powered Smart Public Transport Assistant Using Machine Learning, RAG & LLM

Presented By

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Introduction

Many passengers face difficulties while using public transport due to delayed buses, route confusion, and lack of real-time information.

NEXORA is an AI-powered transport assistant that helps users with live bus tracking, route guidance, **bus arrival prediction**, and instant travel assistance using Artificial Intelligence.



Problem Statement & Objectives

Problem Statement

- No real-time bus tracking
- Long waiting time
- Inaccurate bus arrival prediction
- Route confusion
- Limited passenger guidance

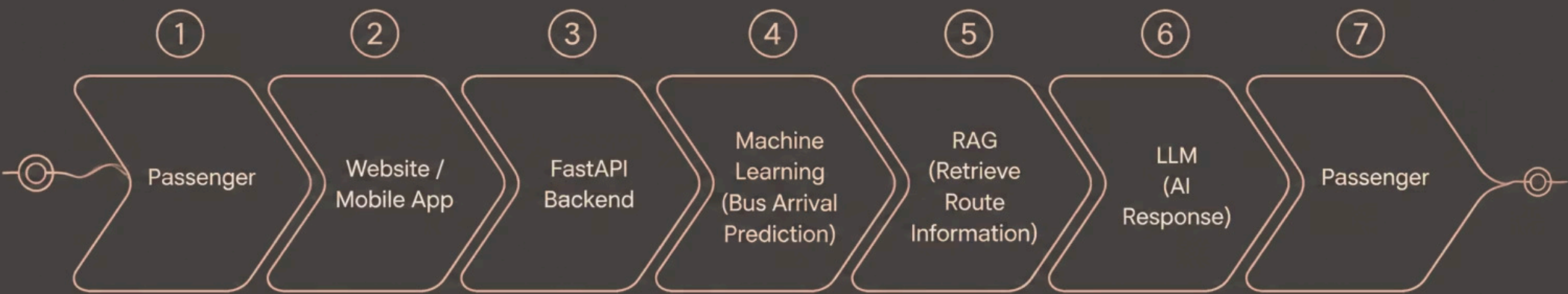
Objectives

- Provide live bus tracking
- Predict bus arrival time
- Suggest the best travel route
- Answer passenger queries using AI
- **Provide women's safety through optional live location sharing and SOS alerts to nearby police stations**
- Improve public transport experience

Existing System vs Proposed System

Feature	MTC Chennai	TNSTC	CUMTA	NEXORA
Bus Route Information	✓	✓	✓	✓
Bus Timetable	✓	✓	Limited	✓
Live Bus Tracking	✗	✗	✗	✓
AI Chat Assistant	✗	✗	✗	✓
ML-Based Bus Arrival Prediction	✗	✗	✗	✓
Conversational Route Guidance	✗	✗	✗	✓
Personalized Route Recommendation	✗	✗	✗	✓
Women's Safety (SOS + Optional Live Location Sharing)	✗	✗	✗	✓
Real-Time Passenger Assistance	✗	✗	✗	✓

System Architecture



Module Usage

Machine Learning
Predicts bus arrival time.

RAG
Retrieves route and bus information.

LLM
Answers passenger questions in simple language.

Tools Used

Component	Technology
Frontend	React.js
Backend	FastAPI
Machine Learning	Scikit-learn
LLM	Llama 3
RAG	LangChain + ChromaDB
Database	PostgreSQL
Maps	Leaflet
Charts	Chart.js

Advantages & Future Scope

Advantages

- Live bus tracking
- Accurate bus arrival prediction
- Smart route guidance
- AI-powered chat assistant
- Reduced waiting time
- Better passenger experience

Future Scope

- Women's safety emergency alert with optional live location sharing
- Voice assistant
- Multi-language support
- Crowd prediction
- Traffic optimization

Conclusion

NEXORA combines Machine Learning, RAG, and LLM to provide a smarter public transport experience.

It helps passengers with bus tracking, route guidance, **bus arrival prediction**, and AI-powered travel assistance, making public transportation more efficient and convenient.



THANK YOU

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